

System identification

$n_x = 6$ was read from the singular-value spectrum of the block Hankel matrix (Figure 1): six singular values exceed the relative threshold $10^{-6}\sigma_1$, with the remainder at numerical zero. All models were fitted with $n_x = 6$.

CVA+EM was fitted first. Its noise matrices, estimated jointly with the dynamics on time-series data, yielded $\text{tr}(Q)/\text{tr}(R) = 0.077$, and one-step-ahead Kalman prediction on $N = 50$ held-out trials ($T = 2000$ steps) gave median $R_F^2 = 0.399$ (84% of the noise-floor ceiling; Figure 4). The downstream muscle decoder maps neural activity to four muscle activation rates through band-pass filters tuned to specific frequencies in the decoded latent signal, so a model that misrepresents the frequency response will produce systematically incorrect muscle activations regardless of state-reconstruction quality. CVA+EM's mean magnitude MAE was 0.086 across all $n_u n_y = 32$ channels, concentrated at low frequencies; phase MAE was 49° , distributed approximately uniformly across frequency (Figure 3). Its transfer function reproduced the per-neuron selectivity for u_1 but failed to recover that for u_2 (Figure 5).

ERA reduced the magnitude MAE to 0.017 and phase MAE to 5° , recovering the empirical per-neuron selectivity for both u_1 and u_2 . Its eigenvalue spectrum contains one complex conjugate pair at ± 0.0274 cycles per step ($|z| = 0.951$); CVA+EM's nearest pair lies at ± 0.0026 cycles per step, an order of magnitude lower. One-step-ahead prediction degraded to $R_F^2 = 0.289$ (51% of ceiling). The Yule-Walker noise estimate produced $\text{tr}(Q) = 1666$ against $\text{tr}(R) = 11.4$, a ratio of 146, so the Kalman filter received negligible information from its own state prediction (Figure 2).

Applying EM to ERA-initialised matrices (ERA+EM) corrected the noise calibration without disturbing the eigenvalue structure. The process noise fell to $\text{tr}(Q) = 0.674$ (ratio 0.048); the dominant complex pair shifted from 0.0274 to 0.0246 cycles per step. Observation reconstruction recovered to $R_F^2 = 0.399$, matching CVA+EM (Figure 4); the ordering $\text{ERA} < \text{CVA+EM} \approx \text{ERA+EM}$ was consistent across all individual output channels. Frequency accuracy (magnitude MAE 0.049, phase 23°) remained substantially better than CVA+EM (Figure 3). ERA+EM was adopted.

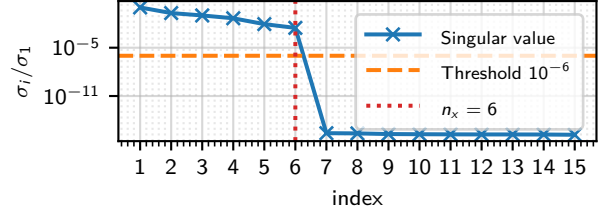


Figure 1: Normalised singular values σ_i/σ_1 of the block Hankel matrix constructed from 55 noise-free Markov parameters (20 block rows, 30 block columns, $n_u = 2$ inputs, $n_y = 16$ outputs). The orange dashed line marks the relative threshold 10^{-6} ; singular values below this level are indistinguishable from numerical zero under exact noise cancellation. The red dotted line indicates $n_x = 6$. [Source](#)

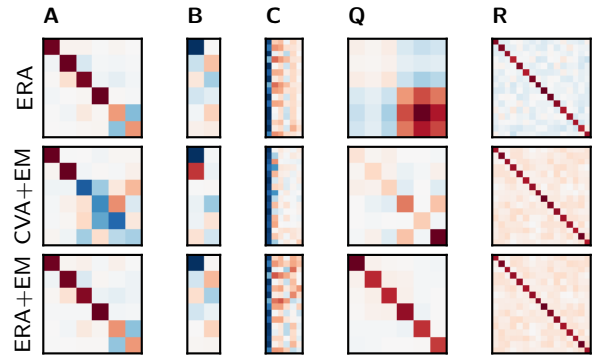


Figure 2: Identified system matrices for ERA, CVA+EM, and ERA+EM (rows, top to bottom). Red indicates positive, blue negative; colour scales are symmetric about zero. [Source](#)

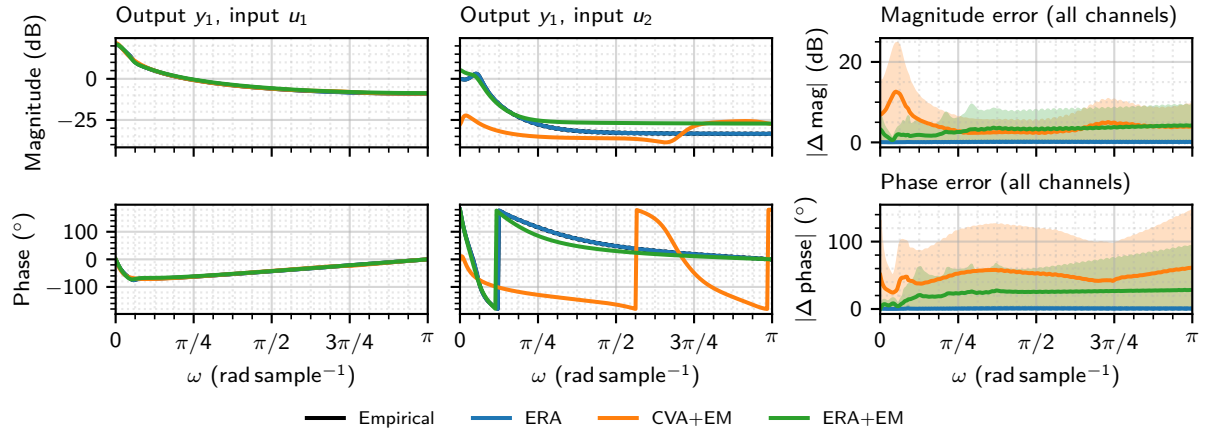


Figure 3: Frequency-response comparison for ERA, CVA+EM, and ERA+EM identified models ($n_x = 6$). Left and centre columns: Bode magnitude (top) and phase (bottom) for output y_1 driven by inputs u_1 and u_2 respectively, comparing the empirical response (discrete Fourier transform of 105 noise-free Markov parameters, zero-padded to 512 points) against the analytic response $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$. Right column: absolute magnitude error (top) and absolute phase error (bottom), shown as mean ± 1 s.d. across all $n_u n_y = 32$ output-input channels for each model. [Source](#)

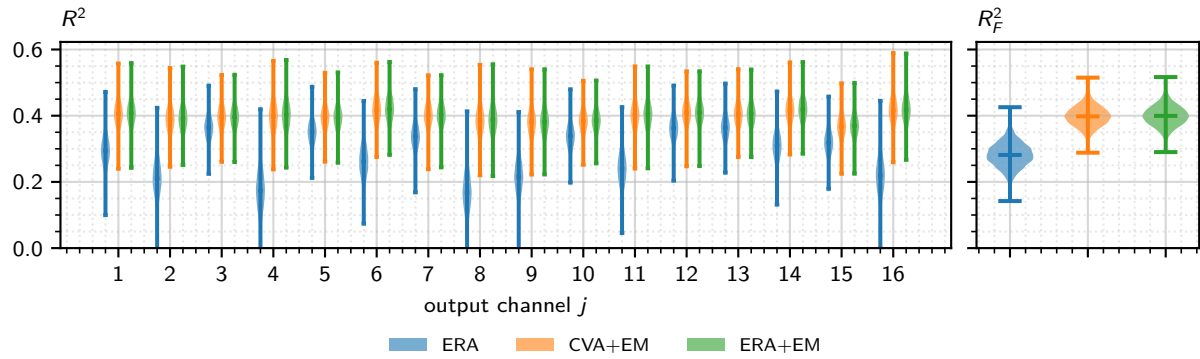


Figure 4: Per-channel R^2 (left, $n_y = 16$) and Frobenius R_F^2 (right) of one-step-ahead Kalman predictions across $N = 10000$ held-out trials of $T = 2000$ steps ($n_x = 6$; first 100 steps excluded). Median: ERA $R_F^2 = 0.281$; CVA+EM $R_F^2 = 0.398$; ERA+EM $R_F^2 = 0.399$. [Source](#)

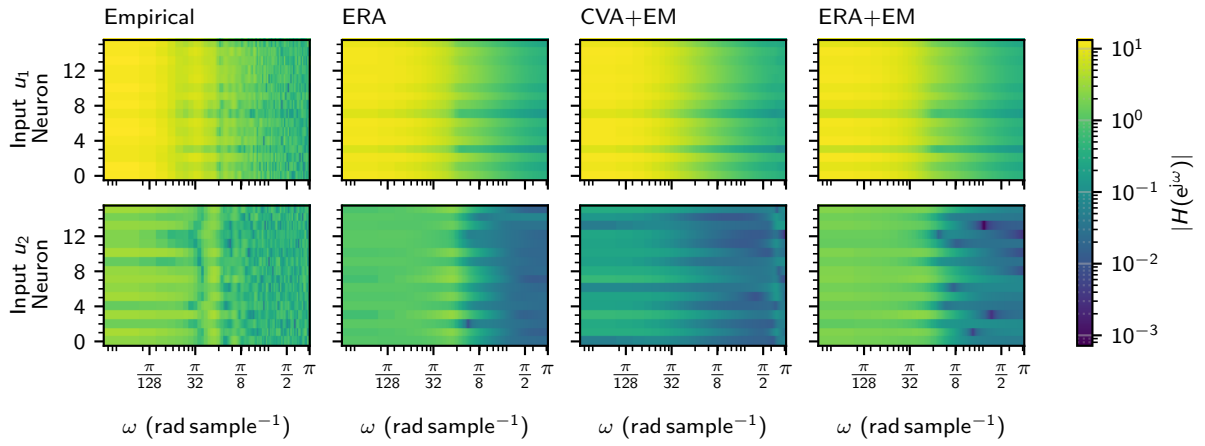


Figure 5: Per-neuron frequency selectivity for inputs u_1 (top row) and u_2 (bottom row). Each panel shows $|H(e^{j\omega})|$ for all $n_y = 16$ neurons (vertical axis) across $\omega \in [0, \pi]$ rad sample $^{-1}$ (horizontal axis), on a shared logarithmic colour scale. Leftmost column shows the empirical response (OLS estimate from a single 5000-step time series, DFT zero-padded to 512 points); remaining columns show the analytic transfer function $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ for ERA, CVA+EM, and ERA+EM respectively. [Source](#)